Q1

$$\nabla f = \left((-2x + 4x^3 y),\ x^4 - 1 \right)$$

$\nabla f = 0 \Rightarrow \quad x = \pm 1 \quad y = 1/2$

The critical points are $(1, 1/2)$, $(-1, 1/2)$

$$D^2 f = \begin{pmatrix} 12x^2y - 2 & 4x^3 \\ 4x^3 & 0 \end{pmatrix}$$

$$D^2 f \Big|_{(-1, 1/2)} = \begin{pmatrix} 4 & -4 \\ -4 & 0 \end{pmatrix} \quad \det < 0 \Rightarrow \text{Saddle}$$

$$D^2 f \Big|_{(1, 1/2)} = \begin{pmatrix} 4 & 4 \\ 4 & 0 \end{pmatrix} \quad \det < 0 \Rightarrow \text{Saddle}$$

The real cps are $(\pm 1, 1/2)$ Both are saddles

Q2

The solution is

$$u(x,t) = \sum_{n=1}^{\infty} e^{-n^2 t} b_n \sin(nx)$$

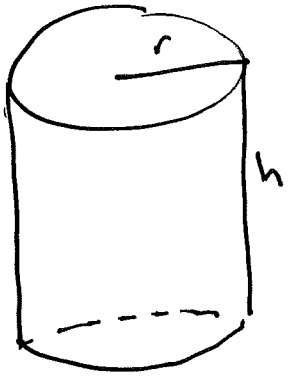
where the b_n's are the Fourier sine coefficients of the initial profile. Since I.C. is already in F.S. form $b_2 = 1$ $b_3 = 1$ $b_i = 0$ for all other i

$$u(x,t) = e^{-4t} \sin(2x) + e^{-9t} \sin(3x)$$

Q3

$$\text{Vol} = \pi r^2 h = 1$$

$$SA = 2\pi r h + 2\pi r^2$$



Need to minimize SA subject to constraint $\text{Vol} = \pi r^2 h = 1$

$$\nabla \text{Vol} = (\pi r^2, 2\pi r h)$$

$$\nabla SA = (2\pi r, 2h\pi + 4\pi r)$$

Need to solve System of 3 eqns

$$\pi r^2 = \lambda \cdot 2\pi r \quad \Rightarrow \quad \lambda = \frac{r}{2}$$

$$2\pi r h = 2h\pi + 4\pi r$$

$$\pi r^2 h = 1 \quad \Rightarrow \quad h = \frac{1}{\pi r^2}$$

Plugging into 2nd eqn

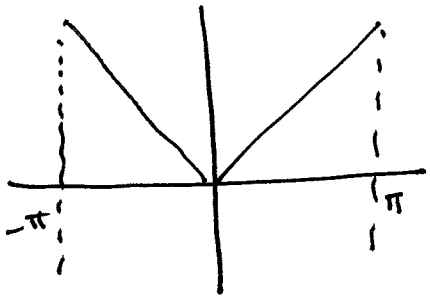
$$\frac{2\cancel{\pi r}}{\cancel{\pi} r^{\cancel{2}}} = \frac{\cancel{r}}{\cancel{2}}\left(\frac{\cancel{2}\pi}{\pi r^{\cancel{2}}} + \overset{2}{\cancel{4}}\pi r\right) \Rightarrow$$

$$\frac{2}{r} = \frac{1}{r} + 2\pi r^2 \quad \Rightarrow \quad 2\pi r^2 = \frac{1}{r} \qquad r = \left(\frac{1}{2\pi}\right)^{1/3}$$

Dimensions are

$$\boxed{r = \left(\frac{1}{2\pi}\right)^{1/3} \text{ cms} \qquad h = \frac{1}{\pi r^2} = \frac{1}{\pi}\left(\frac{1}{2\pi}\right)^{2/3} \text{ cms}}$$

Q4 a)



$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |x|\,dx = \frac{1}{\pi}\int_0^{\pi} x\,dx =$$

$$\frac{x^2}{2\pi}\Big|_0^{\pi} = \frac{\pi}{2}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} |x|\cos(nx)\,dx = \frac{2}{\pi}\int_0^{\pi} x\cos(nx)\,dx =$$

$$\frac{2}{\pi}\left(\frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2}\right)\Big|_0^{\pi} =$$

$$\frac{2}{\pi}\left(\frac{(-1)^n - 1}{n^2}\right) = a_n$$

Fourier series of even periodic extn is

$$\frac{\pi}{2} + \sum_{n=1}^{\infty} a_n \cos(nx)$$

where $a_n = \frac{2}{\pi}\left(\frac{(-1)^n - 1}{n^2}\right)$

b) $|x|$ on $[-\pi, \pi]$ $= \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=0}^{\infty} \frac{\cos((2k+1)x)}{(2k+1)^2}$

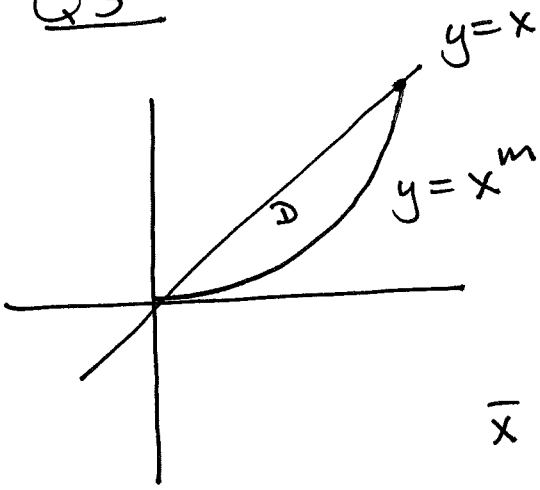
Q 4 (b)

(a) $x=0$ we have

$$\frac{\pi}{2} = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \Rightarrow$$

$$\boxed{\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}}$$

Q5



$$\rho = x+y$$

$$\bar{x} = \frac{\iint_D x\,\rho(x,y)\,dx\,dy}{\iint_D \rho(x,y)\,dx\,dy}$$

$$\bar{y} = \frac{\iint_D y\,\rho(x,y)\,dx\,dy}{\iint_D \rho(x,y)\,dx\,dy}$$

$$\iint_D \rho(x,y)\,dx\,dy = \int_0^1 \int_{x^m}^{x} x+y \; dy\,dx =$$

$$\int_0^1 xy + \frac{y^2}{2}\Big|_{x^m}^{x} = \int_0^1 \frac{3x^2}{2} - x^{m+1} - \frac{x^{2m}}{2}\,dx$$

$$= \frac{x^3}{2} - \frac{x^{m+2}}{2+m} - \frac{x^{2m+1}}{2(2m+1)}\Big|_0^1 = \frac{1}{2} - \frac{1}{m+2} - \frac{1}{2(2m+1)}$$

Q5

$$\int_0^1 \int_{x^m}^{x} (x^2 + xy)\,dy\,dx = \int_0^1 x^2 y + \frac{xy^2}{2} \Big|_{x^m}^{x}$$

$$= \int_0^1 \frac{3x^2}{2} - x^{m+2} - \frac{x^{2m+1}}{2}\,dx$$

$$= \frac{3}{8}x^4 - \frac{x^{m+3}}{m+3} - \frac{x^{2m+2}}{2(2m+2)} \Big|_0^1$$

$$= \frac{3}{8} - \frac{1}{m+3} - \frac{1}{2(2m+2)}$$

$$\int_0^1 \int_{x^m}^{x} xy + y^2\,dy\,dx = \int_0^1 \frac{xy^2}{2} + \frac{y^3}{3} \Big|_{x^m}^{x}$$

$$= \int_0^1 \frac{5x^3}{6} - \frac{x^{2m+1}}{2} - \frac{x^{3m}}{3}\,dx$$

$$= \frac{5}{24}x^4 - \frac{x^{2m+2}}{2(2m+2)} - \frac{x^{3m+1}}{3(3m+1)} \Big|_0^1$$

$$= \frac{5}{24} - \frac{1}{2(2m+2)} - \frac{1}{3(3m+1)}$$

Q5

a) So Centre of mass is @ point

$$\bar{x} = \frac{3/8 - \frac{1}{m+3} - \frac{1}{2(2m+2)}}{1/2 - \frac{1}{m+2} - \frac{1}{2(2m+1)}}$$

$$\bar{y} = \frac{5/24 - \frac{1}{2(2m+2)} - \frac{1}{3(3m+1)}}{1/2 - \frac{1}{m+2} - \frac{1}{2(2m+1)}}$$

b) $$\lim_{m \to \infty} \bar{x} = \frac{3/8}{1/2} = \frac{3}{4}$$

$$\lim_{m \to \infty} \bar{y} = \frac{5/24}{\frac{1}{2}} = \frac{5}{12}$$

Q6

$y_p = A\cos x + B\sin x$

$y_p' = -A\sin x + B\cos x$

$y_p'' = -A\cos x - B\sin x$

$y_p'' + 2y' + y =$

$-A\cos x - B\sin x - 2A\sin x + 2B\cos x + A\cos x + B\sin x$

$= 2B\cos x - 2A\sin x = 2\cos x - 2\sin x$

$\Rightarrow A = B = 1$

homogeneous part $\quad y'' + 2y' + y = 0$

Char eqn $\quad r^2 + 2r + 1 = 0 \Rightarrow (r+1)^2 = 0 \Rightarrow$

$r = -1$

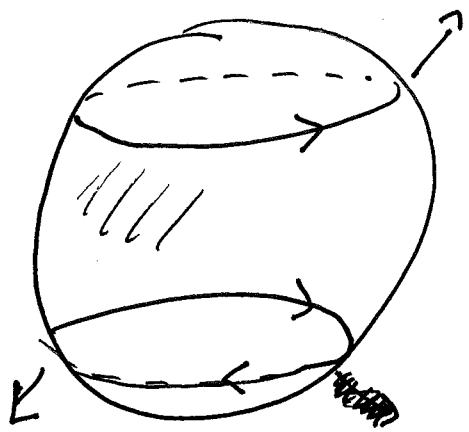
Soln is $\quad y = e^{-x}(c_1 + xc_2)$

So general solution ~~si~~ is $\quad y(x) =$

$$\boxed{e^{-x}(c_1 + xc_2) + \cos x + \sin x}$$

Q7

a)



∂S is two circles w/ opp orientation

$\gamma_1 = (\cos t, \sin t, 1)$

$\gamma_2 = (\cos t, -\sin t, 1)$

b) $F = (yz + 2x, -xz + 4y^2, z^4 - 2)$

$$\iint_S \nabla \times F \cdot n \, dS = \int_{\partial S} F(r)\,dr = \int_{\gamma_1} F(r)\,dr + \int_{\gamma_2} F(r)\,dr$$

$F(\gamma_1) = (2\cos t + \sin t, -\cos t + 4\sin^2 t, -1)$

$F(\gamma_2) = (2\cos t - \sin t, -\cos t + 4\sin^2 t, -1)$

$\frac{d\gamma_1}{dt} = (-\sin t, \cos t, 0)$ $\quad \frac{d\gamma_2}{dt} = (-\sin t, -\cos t, 0)$

$$\int_{\gamma_1} F(r)\,dr + \int_{\gamma_2} F(r)\,dr =$$

Q7 b)

$$\int_0^{2\pi} F(\gamma_1)\frac{d\gamma_1}{dt}\cdot dt + \int_0^{2\pi} F(\gamma_2)\frac{d\gamma_2}{dt}\cdot dt$$

$$= \int_0^{2\pi} -\cancel{\sin^2 t} - 2\cos t\sin t - \cancel{\cos^2 t} + 4\cancel{\cos t}\sin^2 t + \cancel{\sin^2 t} - 2\cos t\sin t + \cancel{\cos^2 t} - 4\cancel{\cos t}\sin^2 t \, dt$$

$$= \int_0^{2\pi} -4\cos(t)\sin t \, dt = \int_0^{2\pi} -2\sin(2t)\, dt = \boxed{0}$$

$\boxed{Q8}$

Solution is $\frac{1}{2c}\int_{x-ct}^{x+ct} \tilde{g}(s)\,ds$ where

$\tilde{g}(s)$ is the Fourier sine series of $x(1-x)$ on $[0,1]$

$c = 3$

$n = 2k+1$

$$b_n = \frac{2}{1}\cdot\int_0^1 x(1-x)\sin(n\pi x)\,dx = \frac{8}{\pi^3(2k+1)^3}$$

$$\text{So } \tilde{g} = \frac{8}{\pi^3}\sum \frac{\sin((2k+1)\pi x)}{(2k+1)^3}$$

$$\int g = \frac{-8}{\pi^3}\sum \frac{\cos((2k+1)\pi x)}{(2k+1)^4\,\pi} = \frac{-8}{\pi^4}\sum_{k=0} \frac{\cos((2k+1)\pi x)}{(2k+1)^4}$$

$$\int_{x-3t}^{x+3t} =$$

$$-\frac{8}{\pi^4}\left(\sum_{k=0}^{\infty} \frac{\cos((2k+1)\pi(x+3t)) - \cos((2k+1)\pi(x-3t))}{(2k+1)^4}\right.$$

Q9

$$\text{Area} = \iint_D \sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2}\, dx\,dy$$

$z = xy \qquad D = x^2+y^2 \le 1$

$$\text{Area} = \iint_D \sqrt{1+y^2+x^2}\, dx\,dy$$

Polar coords

$x = r\cos\theta \qquad D \to [0,1]\times[0,2\pi]$

$y = r\sin\theta$

$dx\,dy \longrightarrow r\,dr\,d\theta$

$$\text{Area} = \int_0^{2\pi}\int_0^1 \left(\sqrt{1+r^2}\right) r\,dr\,d\theta$$

$$= 2\pi \int_0^1 r\sqrt{1+r^2}\, dr$$

$$= 2\pi \int_1^2 r\, u^{1/2}\, \frac{du}{2r} = \pi \int_1^2 u^{1/2}\, du$$

$u = r^2+1 \qquad 0 \to 1$

$du = 2r\,dr \qquad 1 \to 2$

$dr = \frac{du}{2r}$

$$= \frac{2\pi}{3} u^{3/2}\Big|_1^2 = \boxed{\frac{2\pi}{3}(\sqrt{8}-1)}$$